Tracking the Evolution of E-Commerce Websites in Portugal

Exploração de Dados

3rd Delivery

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Context



Purpose

- Study how e-commerce websites in Portugal have changed.



Data

- Gather product information from *Arquivo.pt*.



Objectives

- Analyze changes in product availability, pricing, and diversity over time.

Data Cleaning



Objective

- Clean scraped product data to remove irrelevant or unrelated products.



Why is this necessary?

- The scraped data included products such as: chargers, cables, covers, books, TVs, etcs.
- Uncleaned data will affects accuracy in product prices analysis.

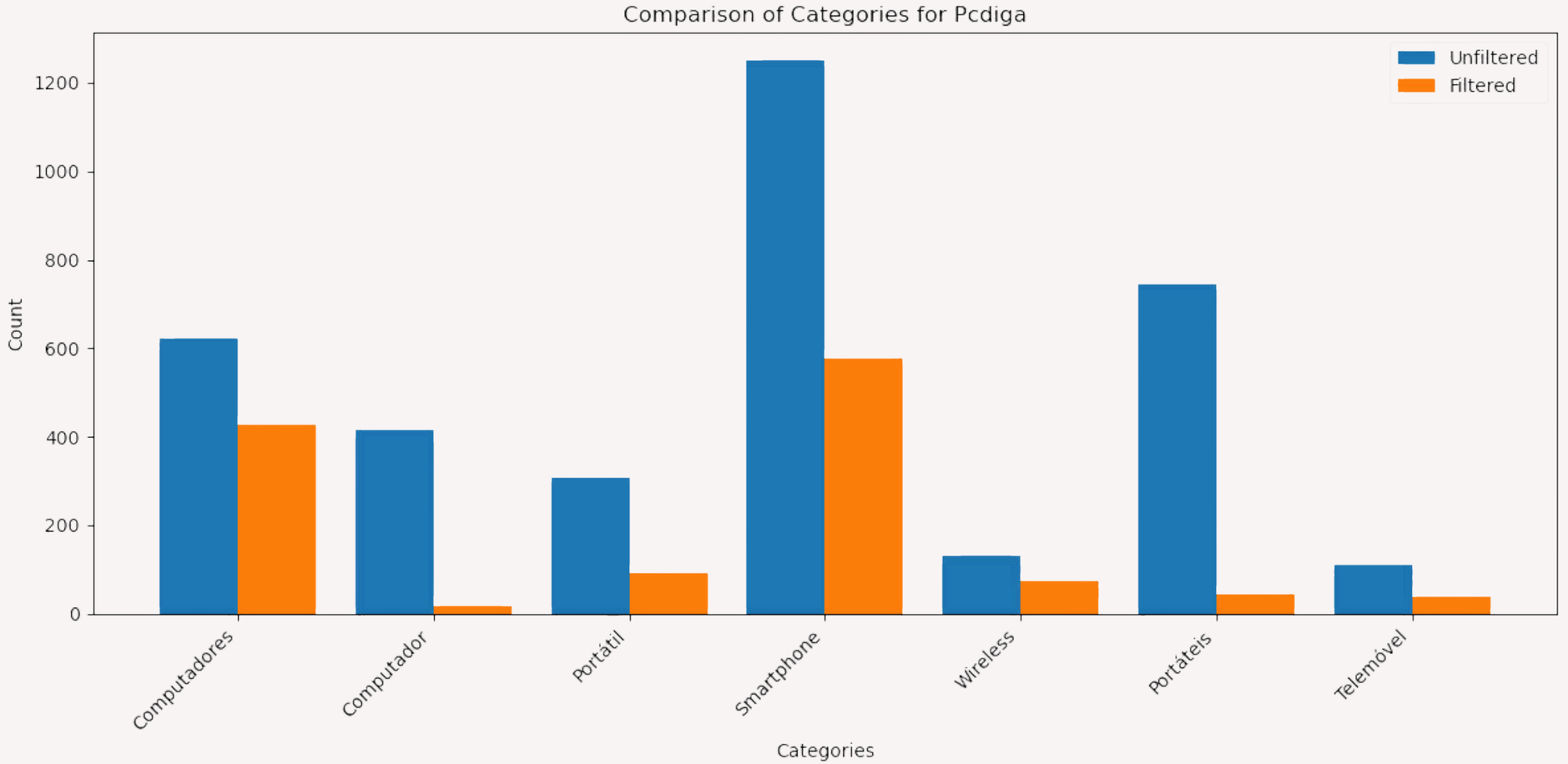
Data Cleaning



Implementation

- For each category we defined a list containing keywords that indicate unrelated products from the category, like **Telemóvel: “capa”, “headset”, “bluetooth”, “tv”**.
- We filter the data by removing rows with that keywords in the title.
- Store results in a new CSV files for further analysis.

Data Cleaning - Results



- The filtered data has approximately 56% fewer items compared to the unfiltered data.

Price Anomaly Detection



Approach

- We use **Isolation Forest**, an unsupervised machine learning model.



Objective

- Identify **unusual price changes** in product data using a robust anomaly detection method.



Isolation Forest

- Isolates data points by randomly splitting features.
- Anomalies are isolated faster.

Isolation Forest



Implementation

- Input data: Product titles, timestamps, and prices.



Resample Data

- Resample prices monthly using the mean.



Anomaly Detection

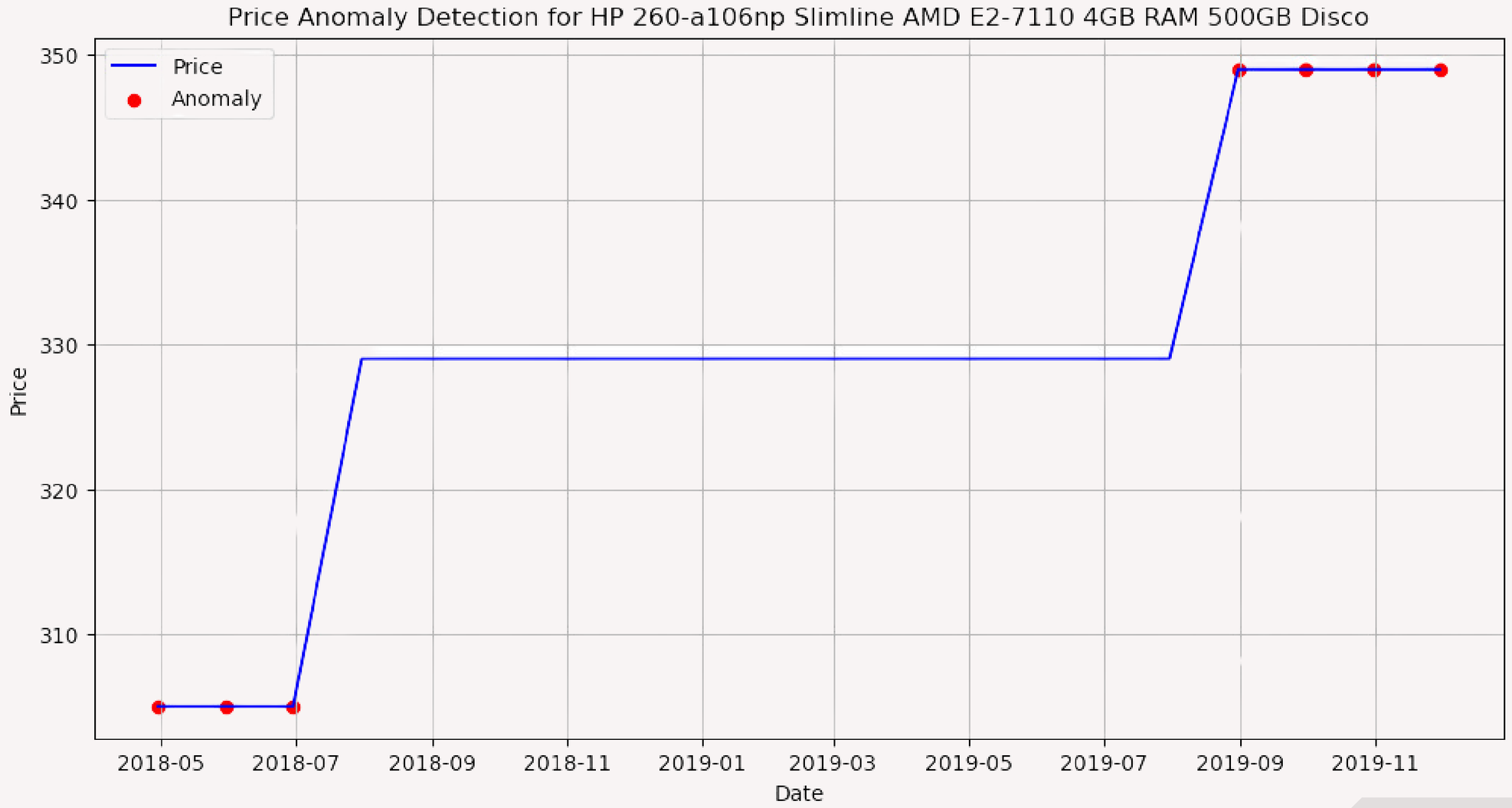
- Train the model on the product’s price data.



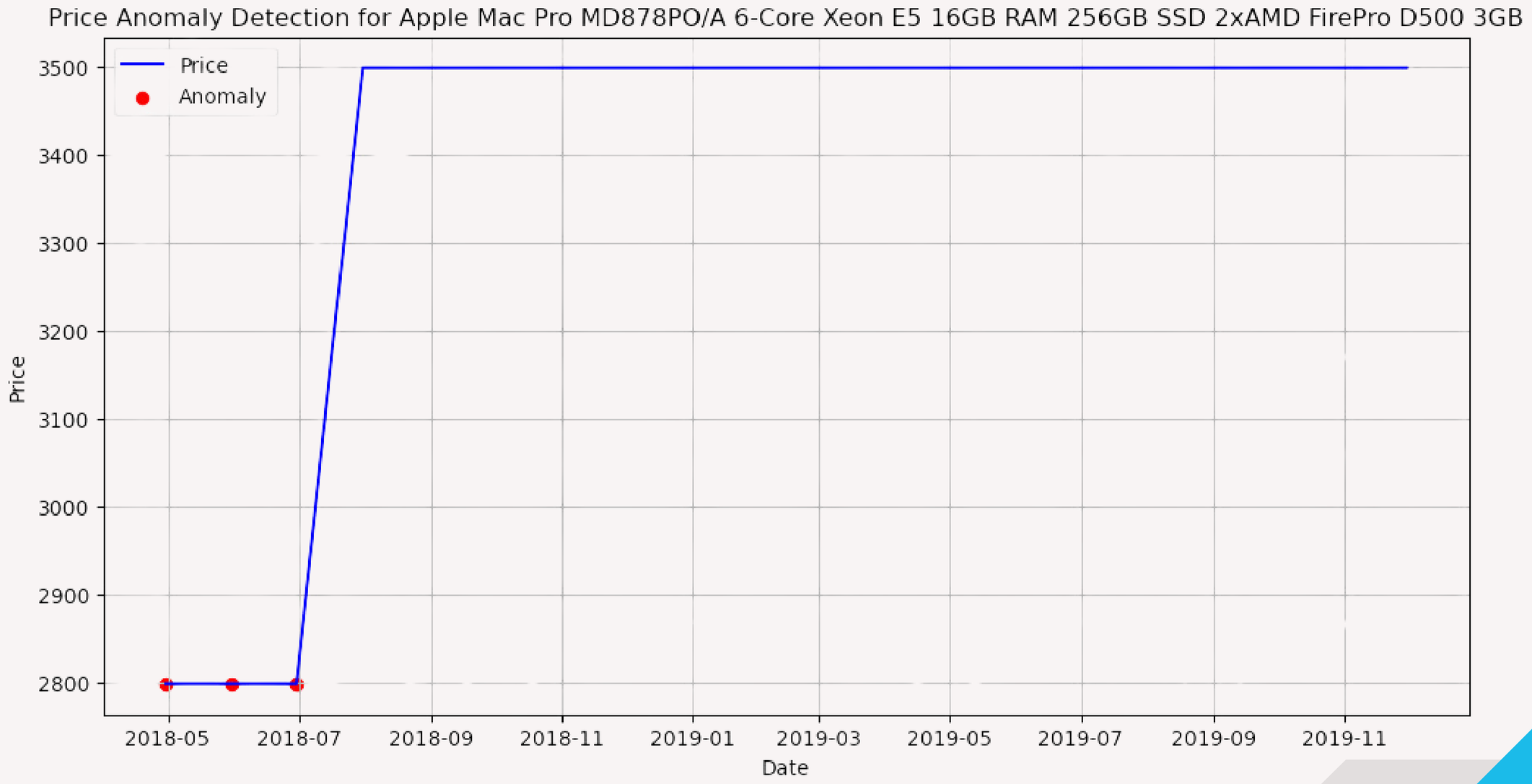
Visualuzations

- Plot price trends with detected anomalies.

Price Anomaly Detection



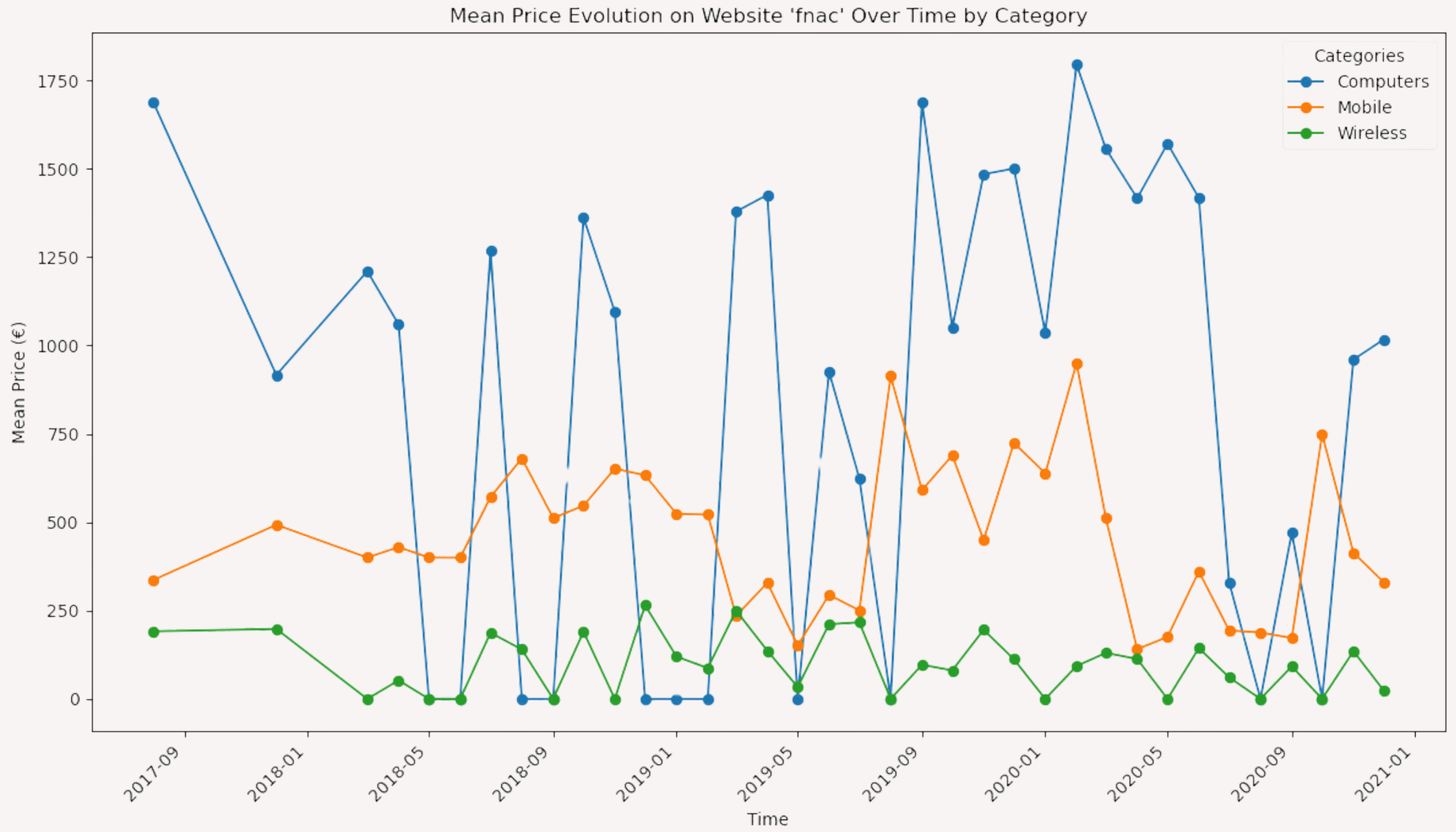
Price Anomaly Detection



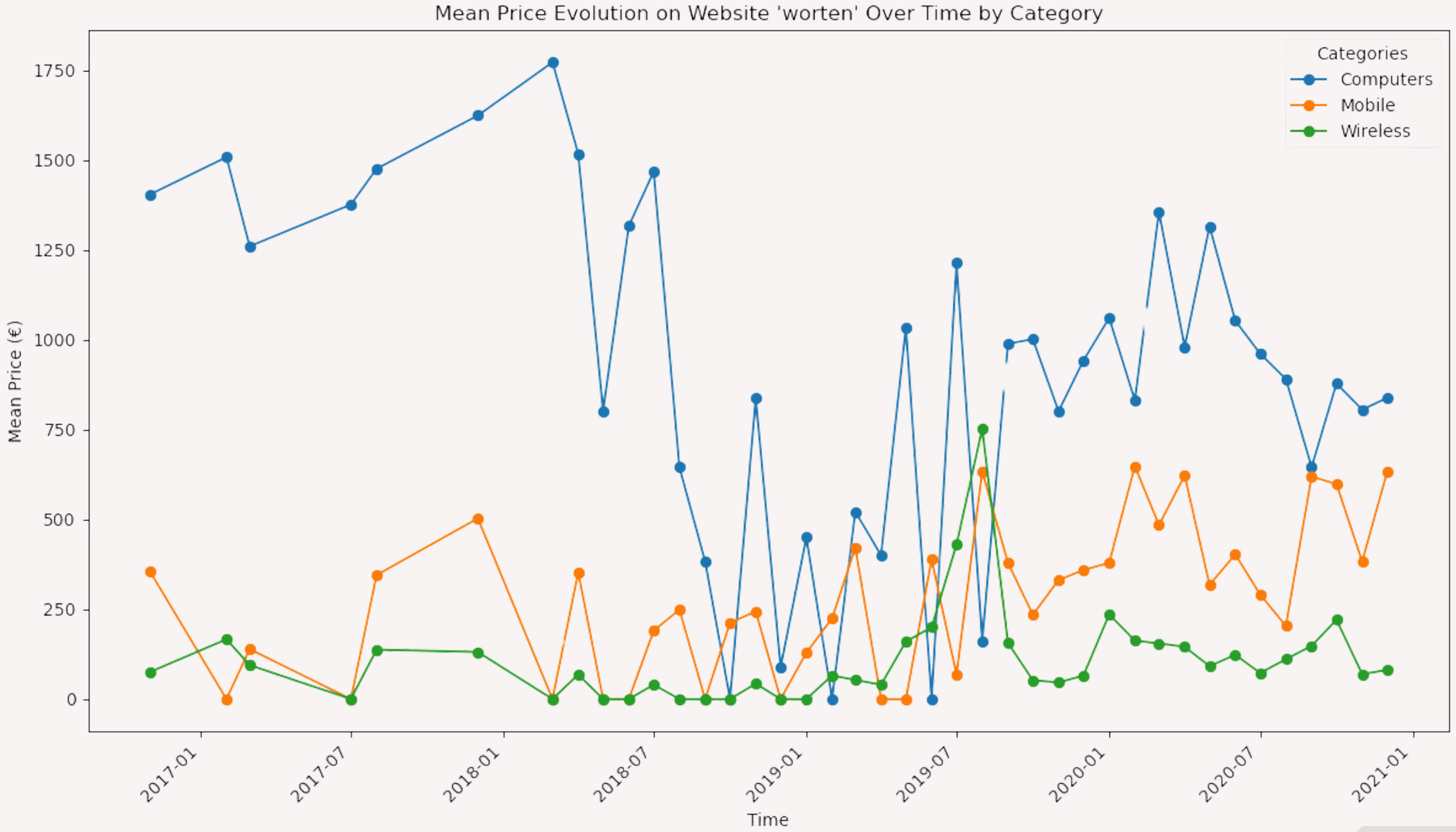
Price Evolution

Categories

Price Evolution - Categories



Price Evolution - Categories



Price Evolution - Normalization



Problem

- Product titles vary slightly across websites, so we needed to group identical products.



Solution

- Cleaned product titles and grouped similar ones using fuzzy matching.



Tool Used

- Fuzzy Matching
  Python Library - RapidFuzz

Price Evolution - Especific Product



Price Evolution - Especific Product



User Interface

Possible Improvements

- **Anomaly detection** can be enhanced by combining multiple models like **Isolation Forest**, **LOF**, and **DBSCAN**. This ensemble approach ensures more accurate and robust detection by reducing reliance on a single model.

- **Product title** matching can be improved using **NLP** embeddings (e.g., **Word2Vec**) to cluster similar titles. This method better identifies duplicates by analyzing context, leading to more precise product comparisons and trend analysis.

THANKS!

Do you have any questions?

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